

ADVANCED DEFENSE ELECTRONICS

BIST EMBEDDED TELEMETRY BOARD

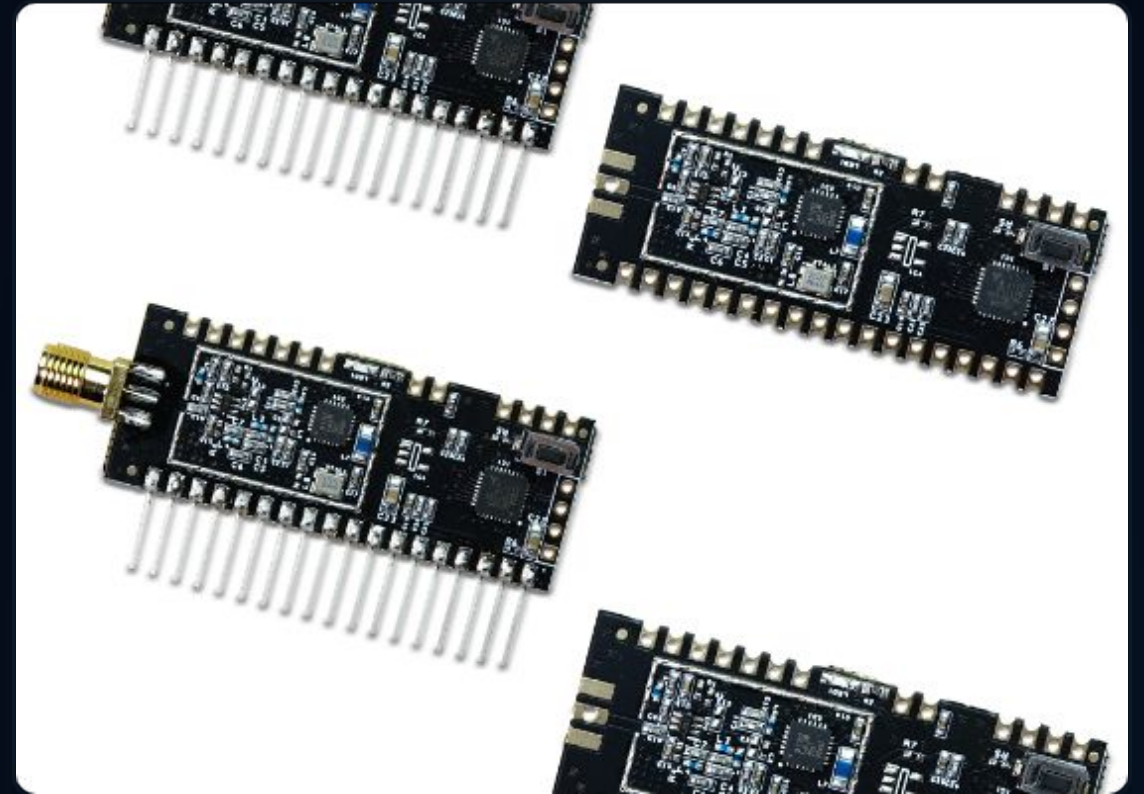
A Comprehensive Design for Reliability, Fault-Tolerance, and High-Speed
Data Acquisition in Mission-Critical Systems.

OVERVIEW OF TELEMETRY SYSTEMS



Telemetry involves the automatic measurement and wireless transmission of data from remote sources to IT systems. In the context of **Bharat Electronics Limited (BEL)**, telemetry is vital for:

- ✓ **Missile Health Monitoring** during flight.
- ✓ **UAV Control Systems** & Flight Data.
- ✓ **Radar Diagnostics** & Remote Sensing.
- ✓ **Satellite Communication** Link Metrics.



| THE ROLE OF BUILT-IN SELF-TEST



AUTONOMOUS TESTING

BIST allows the hardware to test itself without requiring external Automatic Test Equipment (ATE).



HIGH RELIABILITY

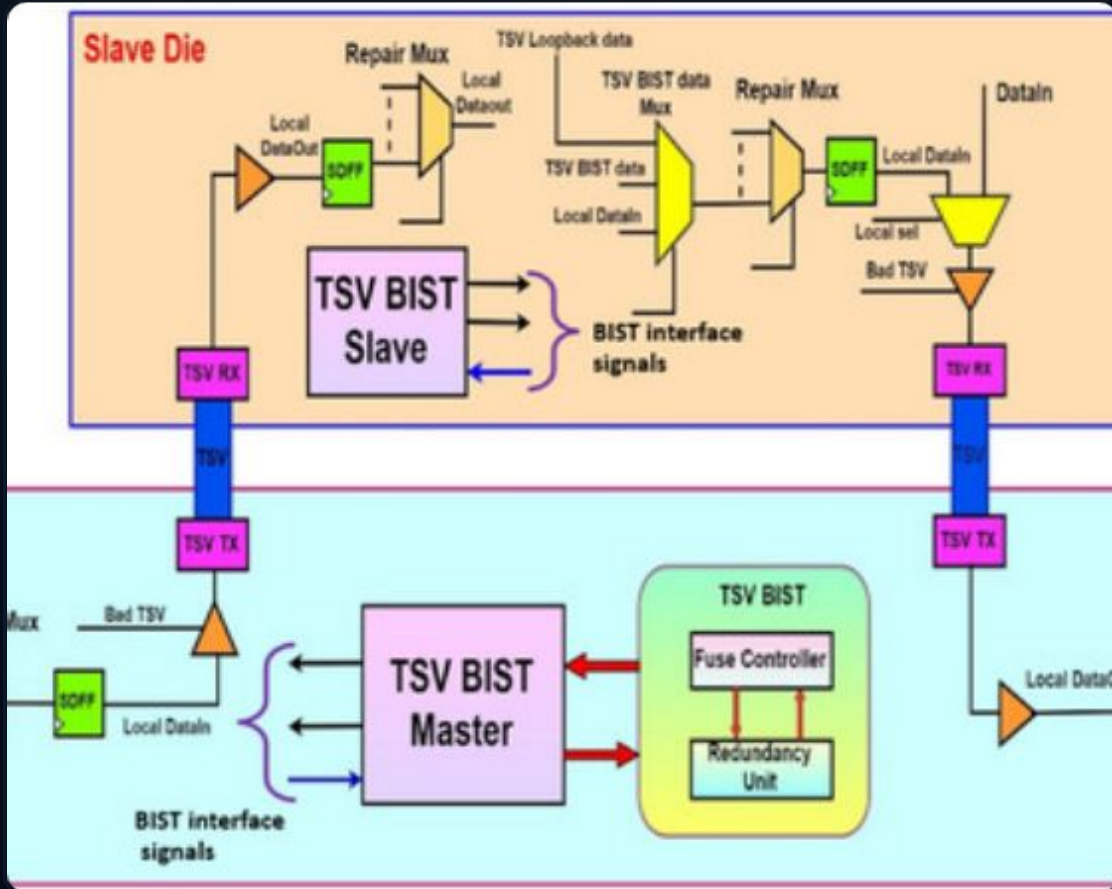
Ensures the telemetry board is fully functional before and during critical mission phases.



REDUCED DOWNTIME

Rapidly isolates faults down to the component level, simplifying maintenance and repair.

BOARD SYSTEM ARCHITECTURE

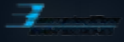


UNIFIED DESIGN FLOW

The architecture integrates data processing paths with parallel BIST logic. This allows for concurrent monitoring of system health while processing real-time sensor data.

$$\text{Reliability} = e^{-\lambda t}$$

Where λ is the failure rate and t is the operation time.



FIELD PROGRAMMABLE GATE ARRAY

Definition: An integrated circuit designed to be configured by a customer or a designer after manufacturing.

Usage:

- ✓ Implements BIST test pattern generators.
- ✓ Handles high-speed signal processing.
- ✓ Manages real-time data encryption.

CENTRAL MICROCONTROLLER (MCU)

SYSTEM ORCHESTRATION

The MCU acts as the 'brain' for high-level logic and communication management.

- ✓ **Control:** Triggers the BIST sequences.
- ✓ **Storage:** Manages error log generation.
- ✓ **Interface:** Communicates with the ground station via standard protocols.



Firmware Logic

Implements RTOS for deterministic telemetry task execution.

| ANALOG TO DIGITAL CONVERTERS



SIGNAL CONVERSION

Transforms raw analog signals from sensors into 16-bit or 24-bit digital packets.

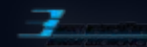


HIGH SAMPLING RATE

Capable of sampling at MSPS rates to capture transient flight events.

BIST Usage: Internal calibration loops verify ADC accuracy during the power-on self-test (POST).

| TELEMETRY SENSOR SUITE



The telemetry board interfaces with a variety of MEMS and high-precision sensors:

IMU

Accelerometers & Gyroscopes for orientation.

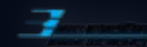
THERMISTOR

Monitoring critical component temperature.

PRESSURE

Altimeter data for altitude tracking.

MEMORY & DATA STORAGE



NON-VOLATILE FLASH

Used for "Black Box" logging of flight parameters. Data is preserved even in the event of power loss.

$$\text{Capacity} = \text{BitRate} \times \text{Duration}$$

DDR4 SDRAM

High-speed buffer for FPGA processing and temporary data staging before RF transmission.

| RF & SERIAL TRANSCEIVERS



RF TRANSMITTER

UHF/S-Band modules for transmitting telemetry packets to ground stations.



RS-422 / LVDS

Robust differential signaling for internal hardware communication in noisy environments.

| BIST BLOCK: TEST PATTERN GENERATOR

The TPG generates pseudo-random test vectors using **Linear Feedback Shift Registers (LFSR)**.

- ✓ Exhaustive coverage of internal logic.
- ✓ Deterministic yet pseudo-random.
- ✓ Minimized hardware overhead.

LFSR Polynomial:

$$P(x) = 1 + x^3 + x^7$$

| BIST BLOCK: OUTPUT RESPONSE ANALYZER



The ORA compresses the output response from the Circuit Under Test into a unique **Signature**.

MISR TECHNIQUE

Multiple Input Signature Register compacts multi-bit outputs into a single hex code.

GOLDEN SIGNATURE

The result is compared against a hard-coded 'Golden Signature' to determine PASS/FAIL status.

| BIST CONTROLLER LOGIC

STATE MACHINE

- Idle State
- Init Pattern Gen
- Shift Data into Scan Chains
- Capture Response
- Compare & Report

The controller manages the timing and synchronization between the TPG and ORA. It ensures that the test doesn't interfere with the telemetry board's functional operation during "In-Circuit" testing.

OPERATIONAL DATA FLOW

SENSING

Raw Data

ADC CONV

Digital Samples

FPGA PROC

Filtered Packets

RF TRANSMIT

Wireless Uplink

| TEST COVERAGE & METRICS

99.8%

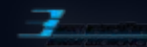
FAULT COVERAGE

RELIABILITY STANDARDS

The BIST implementation is designed to meet MIL-STD-883 standards for defense electronics.

- ✓ Stuck-at Fault Detection.
- ✓ Bridging Fault Analysis.
- ✓ Delay Fault Monitoring.

| POWER MANAGEMENT UNIT



MULTI-RAIL LDOs

Providing clean 1.8V, 3.3V, and 5V rails for mixed-signal components.

CURRENT MONITORING

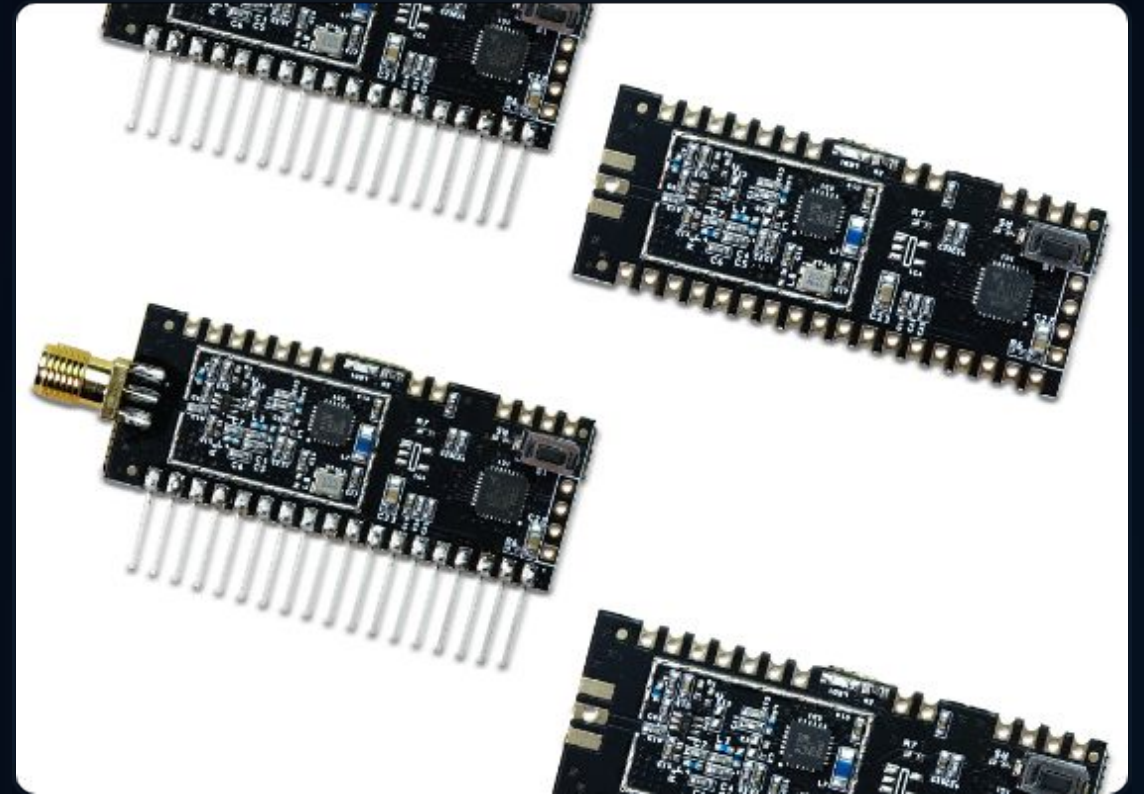
BIST logic monitors current draw to detect internal shorts or thermal runaway.

DEPLOYMENT IN BEL PROJECTS



BEL utilizes these embedded telemetry boards in several flagship systems:

- ✓ **Akash Missile System:** Telemetry for flight trajectory.
- ✓ **Astra BVR:** Real-time health diagnostics.
- ✓ **Coastal Surveillance:** Remote radar node monitoring.



| HARDWARE IMPLEMENTATION CHALLENGES

AREA OVERHEAD

BIST logic can consume up to 15% of FPGA gate area.

EMI/EMC

High-speed switching in BIST circuits must not affect RF sensitivity.

THERMAL MANAGEMENT

Dense telemetry boards require active heat dissipation.

| THE FUTURE OF EMBEDDED BIST

- ✓ **AI-Driven Diagnostics:** Predicting failures before they happen using edge ML.
- ✓ **Adaptive BIST:** Modifying test patterns based on environment stressors.
- ✓ **Chiplet Integration:** Multi-die telemetry solutions for miniaturization.

THANK YOU

ANY QUESTIONS?

Presented by Technical Engineering Team | Bharat Electronics Limited